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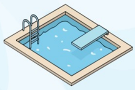
AI-Based Smart Swimming Pool Safety and Drowning Detection System

نظام ذكي قائم على الذكاء الاصطناعي لرصد حالات الغرق في برك السباحة



Introduction

- Drowning is a leading cause of accidental death globally, claiming an estimated 236,000 lives annually.
- It is the #1 cause of injury death for children aged 1–4 years.
- The Core Challenge: Drowning is silent (no splashing) and quick (4–6 minutes to brain damage).
- Human Limitations: Human lifeguards suffer from vigilance decrement (loss of focus) after just 30 minutes due to fatigue, glare, and distractions.

Objectives

- The goal is to engineer an autonomous, "unblinking" Digital Lifeguard.
- Develop a 24/7 continuous monitoring system using low-cost edge computing (no cloud needed).
- Detect dangerous immobility indicative of drowning within seconds.
- Identify unauthorized access by small children approaching the pool deck.
- Alert bystanders instantly with local high-intensity audio and visual warnings.

Core Operational Workflow

The system uses a defined logic loop to differentiate safe swimming from distress.

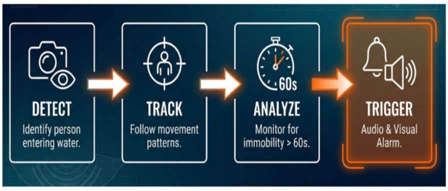


Figure 3 The algorithmic pipeline from detection to alarm trigger.

1. ROI Masking: A user-defined polygon isolates the water area, ignoring the pool deck to prevent false alarms.
2. Velocity Tracking: The system calculates the distance a tracked ID moves between frames.
3. The Critical Threshold: If velocity drops near zero for 60 consecutive seconds, the state changes to "DANGER."

User Interface & Detection Examplesnal

The system provides real-time visual feedback.
The ROI masking screenshot (green polygon).
The final "Suspected Drowning" red box alert screenshot.

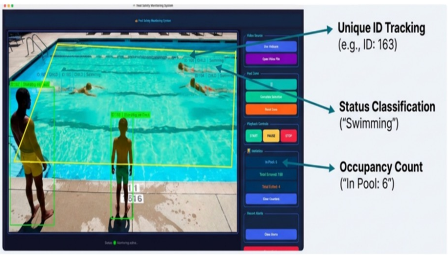


Figure 4: Region of Interest masking ensures focus on the water. The system detected immobility and triggers a visual Green alert.

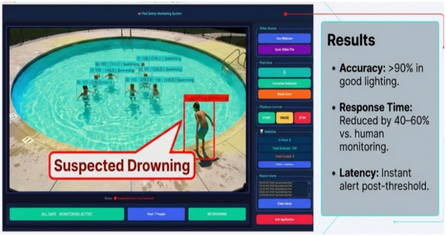


Figure 5 : Region of Interest masking ensures focus on the water. The system detects immobility and triggers a visual red alert.

References

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Our Team

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Methodology

System Hardware

- We utilized an edge-computing approach to ensure speed and privacy.
- Processing Unit: Raspberry Pi 5 (high-performance localized AI processing).
- Vision Sensor: High-Resolution Wide-Angle Camera Module (mounted overhead).
- Output: GPIO-connected active buzzer and visual display interface.
- Housing: Custom 3D-printed, weather-resistant enclosure.

Figure 1: Hardware prototype showing the Raspberry Pi 5 compute unit and custom-designed housing.This Image Created by Team using MLNotebook.

Software Stack & AI Engine

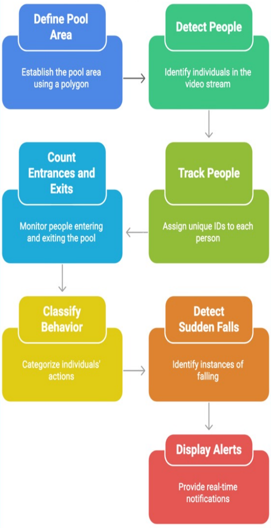
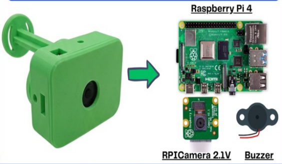


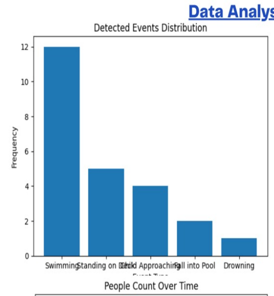
Figure 2: System Methodology Workflow,This Image Created by Team using Napkin.

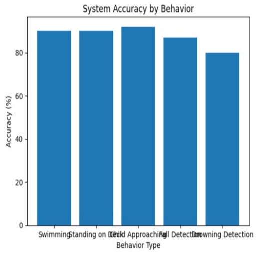


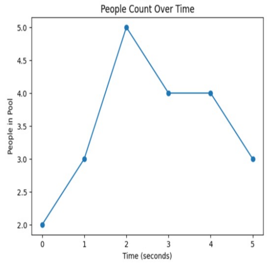
FUTURE WORK

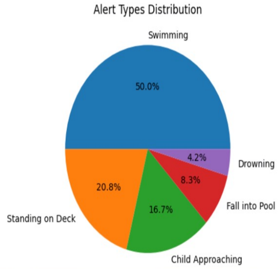
- Mobile app notification
- Cloud storage
- Night vision support
- Sound detection for screams
- Thermal camera integration

Data Analysis & Interpretation









Hardware Demonstration

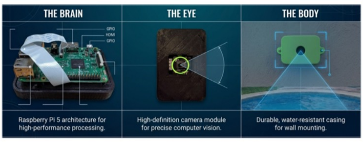


Figure 6: Digital boundaries and ROI (Region of Interest) masking interface.

System Demonstration